EXPERIMENT 8

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

One-Sample Z-Test

Aim: Test whether mean bird weight differs from 150 g at 5% significance.

Python Program

```
import numpy as np
from scipy.stats import norm
data=np.array([152,148,151,149,147,153,150,148,152,149,151,150,149,152,151,148,150,152,149,150,148,153,151,150,149,152,148,151
z=(data.mean()-150)/(data.std(ddof=1)/np.sqrt(len(data)))
p=2*(1-norm.cdf(abs(z)))
print("Mean:",data.mean())
print("Z:",z,"P:",p)
print("Reject H0" if p<0.05 else "Fail to reject H0")
```

Result

For the manual's sample, mean is about 150.20, Z about 0.64 and P about 0.52. Fail to reject H0.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.